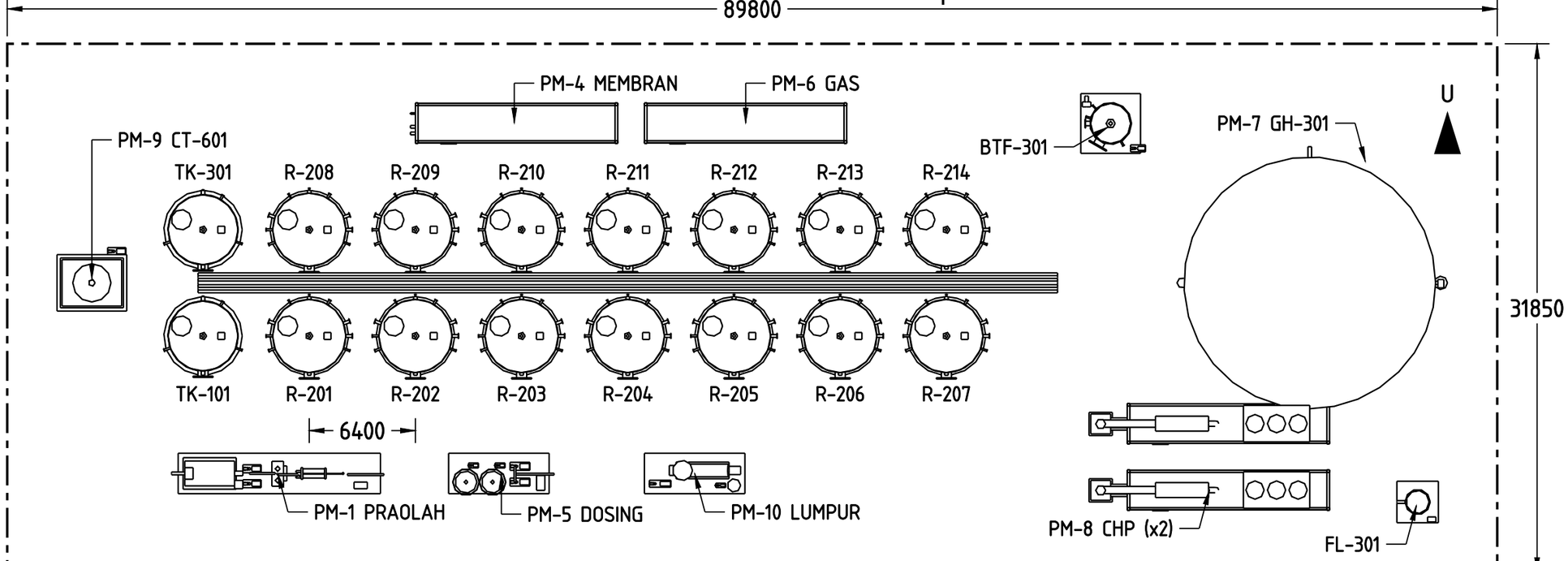
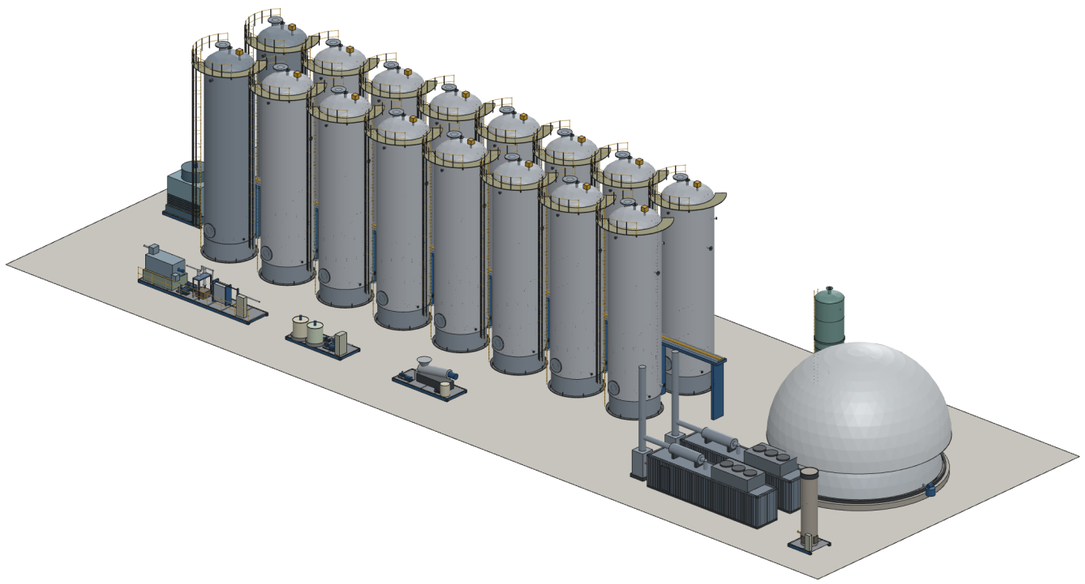


DENAH TATA LETAK (DARI MODEL 3D)

SKALA 1:300



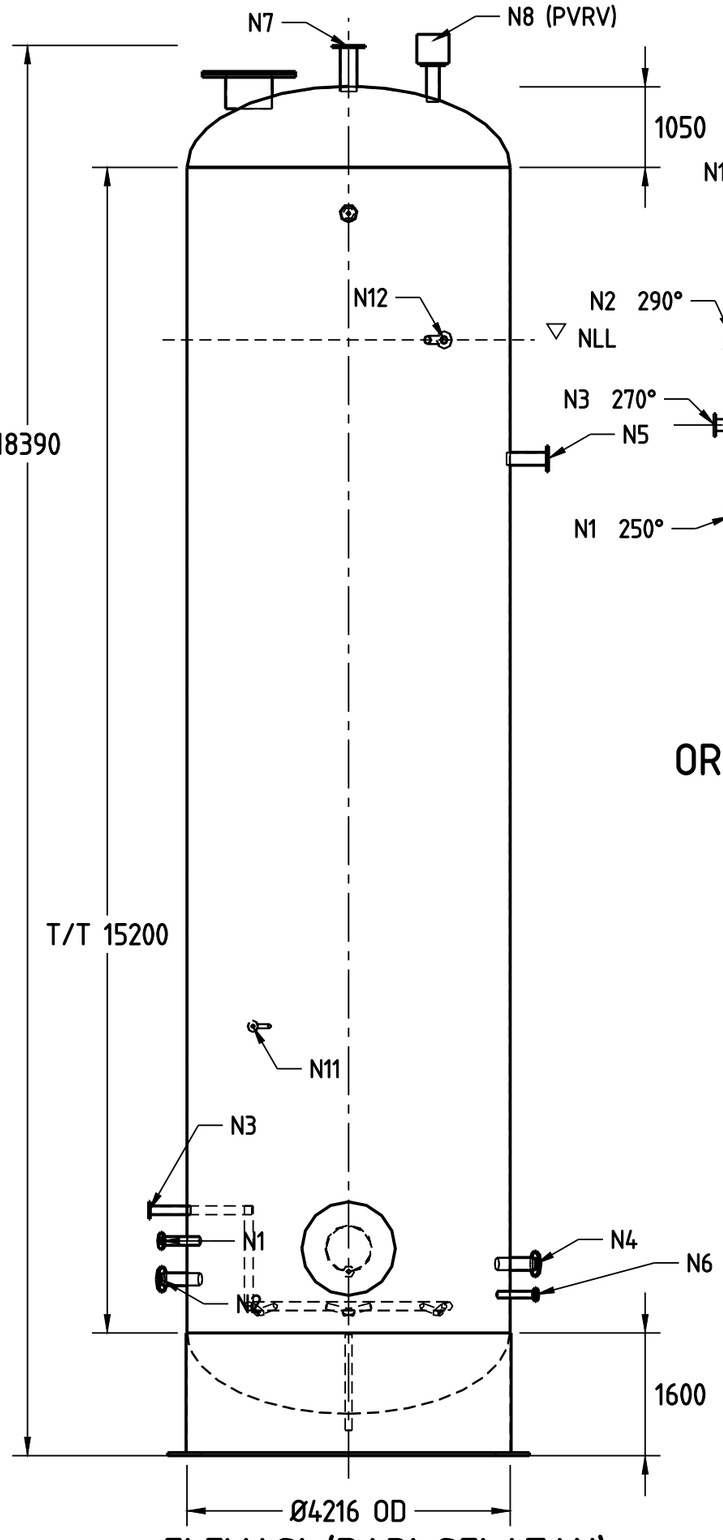
ISOMETRI SUSUNAN PABRIK (TANPA SKALA)

SET MODUL PORTABEL				
Modul	Nama	Jml	Pengiriman	Massa t
PM-1	Skid praolah POME	1	kontainer 40 ft (skid)	12.0
PM-2	Tangki penyangga umpan	1	bejana di trailer low-bed	19.0
PM-3	Reaktor AnMBR termofilik	14	bejana di trailer low-bed	266.0
PM-4	Kontainer membran + CIP	1	kontainer 40 ft HC	16.0
PM-5	Skid dosing dan pompa umpan	1	skid 20 ft	8.0
PM-6	Skid pengolahan biogas	1	kontainer 40 ft + bejana BTF	14.0
PM-7	Penampung gas + flare	1	kontainer 20 ft + skid flare	7.0
PM-8	Kontainer CHP	2	kontainer 40 ft HC	48.0
PM-9	Skid pemulih panas + menara pendingin	1	skid 20 ft + sel CT modular	9.0
PM-10	Tangki permeat + skid lumpur	1	bejana + skid 20 ft	27.0

Total 28 muatan truk, ± 426 t. Plot 89.8 x 31.9 m = 0.29 ha (termasuk jalan 3 m).

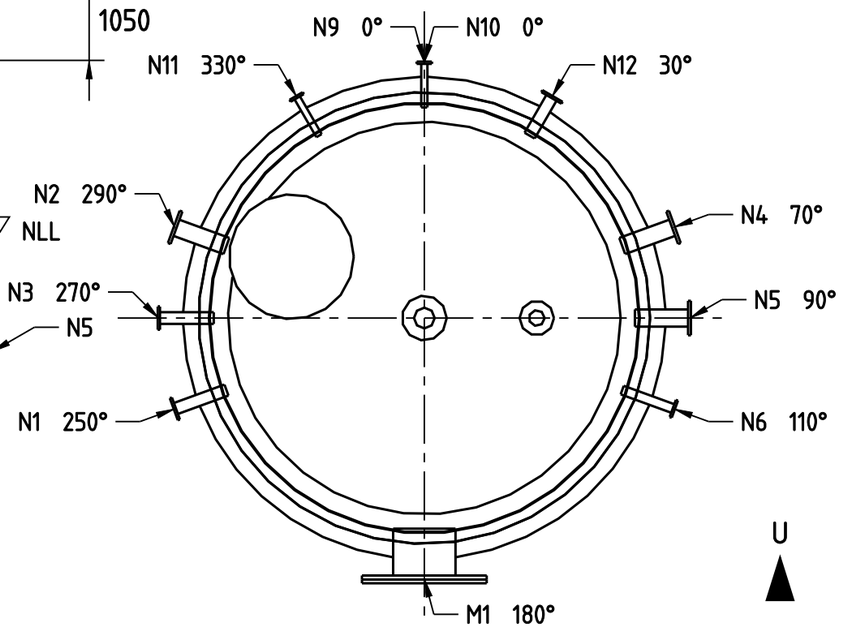
SKALA SET (DESAIN BEJANA SAMA)					
PKS	Reaktor	Genset	Neto kWe	GWh/thn	Membran m²
30 t/jam	10	2 x 600 kWe	627	4.5	634
45 t/jam	14	2 x 800 kWe	961	6.9	996
60 t/jam	19	3 x 600 kWe	1311	9.4	1314

University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: General Arrangement			Rev: 0
	Author : NAMA TIM	Tata Letak Umum Set Portabel PKS 45 t TBS/jam	Unit: mm	Scale: 1 : 300	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-GA-001			Sheet number: 1 / 8



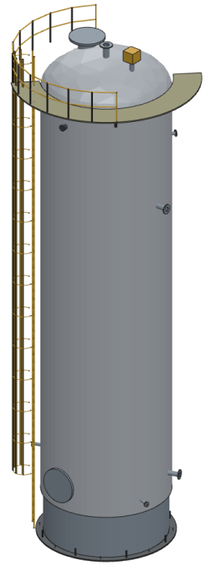
ELEVASI (DARI SELATAN)

SKALA 1:80



ORIENTASI NOSEL (TAMPAK ATAS)

SKALA 1:60



ISOMETRI (TANPA SKALA)

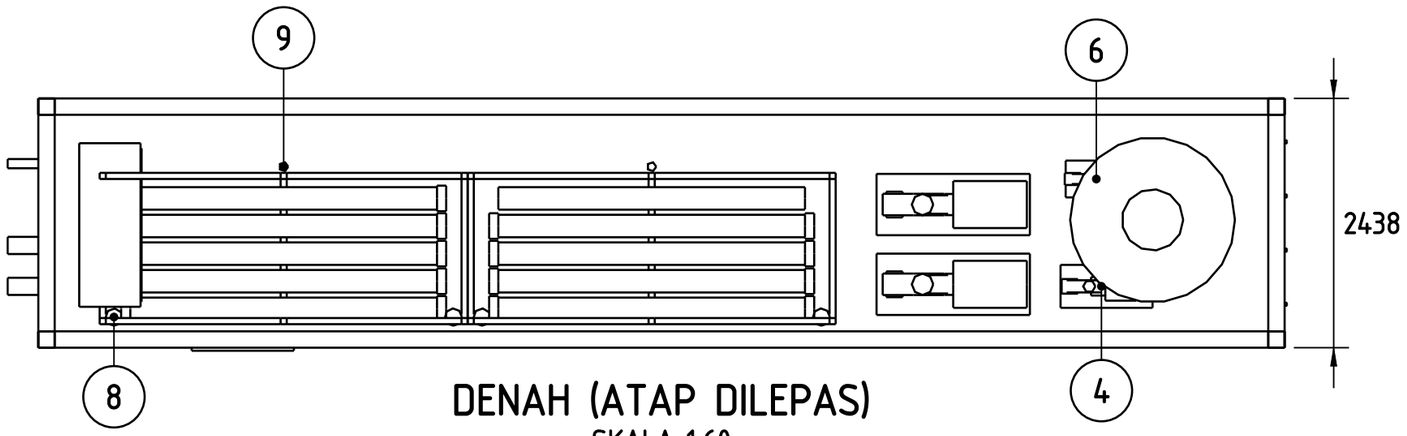
JADWAL NOSEL					
Tag	Ukuran	Rating	Layanan	Elev.	Orient.
N1	DN100	PN10 RF	Umpan reaktor (dari P-201)	2800	250°
N2	DN150	PN10 RF	Isap jet mixing (ke P-202)	2300	290°
N3	DN100	PN10 RF	Balik jet mixing (header nosel jet)	3200	270°
N4	DN150	PN10 RF	Lumpur ke membran (ke P-301)	2500	70°
N5	DN150	PN10 RF	Retentat balik (dari MB-301)	13000	90°
N6	DN80	PN10 RF	Lumpur buangan (ke P-303)	2100	110°
N7	DN200	PN10 RF	Keluaran biogas	puncak	–
N8	DN150	PN10 RF	PVRV / relief tekanan-vakum	kepala atas	–
N9	DN50	PN10 RF	Pemancar level LT (atas / bawah)	16200	0°
N10	DN50	PN10 RF	Pemancar level LT bawah	2400	0°
N11	DN50	PN10 RF	Termometer TT / pH / NH4	5600	330°
N12	DN100	PN10 RF	Kaca intip muka cairan	14550	30°
M1	DN600	PN10 RF	Lubang orang bawah	2700	180°
M2	DN600	PN10 RF	Lubang orang atas	kepala atas	–
N13	DN80	PN10 RF	Pengurasan (kepala bawah)	330	–

DATA DESAIN	
Data desain	Nilai
Fungsi	Reaktor AnMBR termofilik (14 unit identik)
Kode desain	API 650 / ASME VIII-1 (acuan), bejana atmosferik
Tekanan desain	+50 / −5 mbar(g) + hidrostatik penuh
Suhu desain / operasi	70 °C / 55 °C
Volume total / kerja	230 m³ / 194 m³
Bahan shell / kepala	SA-516-70, 8 mm, CA 3 mm, lapisan epoksi
Kepala	Elipsoidal 2:1, 8 mm
Insulasi	Mineral wool 100 mm + kladding Al 0,8 mm
Massa kosong / operasi	± 20 t / ± 218 t
Transport	rebah di low-bed, 17.3 x 4.4 m

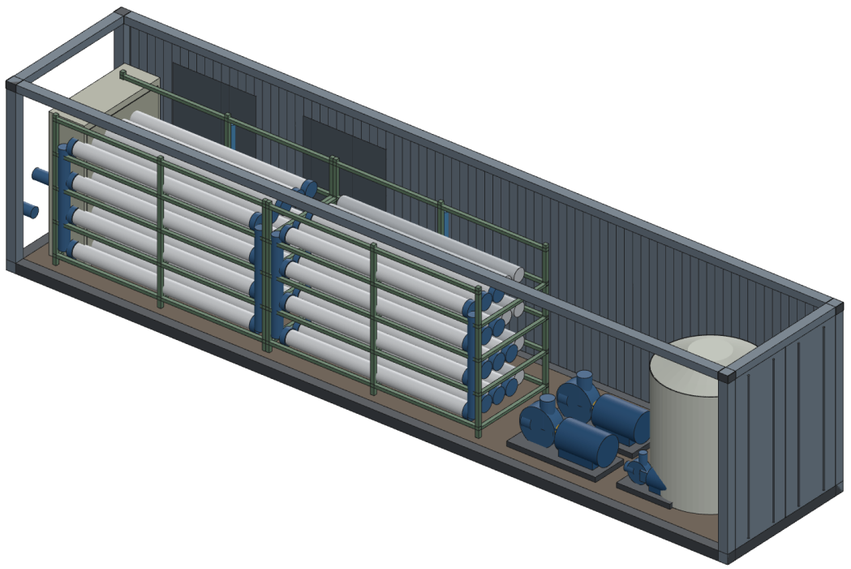
CATATAN :

1. Elevasi diukur dari dasar cincin rok (EL 0).
2. Orientasi nosel: derajat dari utara, searah jarum jam.
3. Tangga, sangkar & platform atas: lihat isometri.
4. Bejana sama dipakai TK-101 / TK-301 (tanpa insulasi & internal).
5. Header jet internal (6 nosel) ditunjukkan garis tersembunyi.

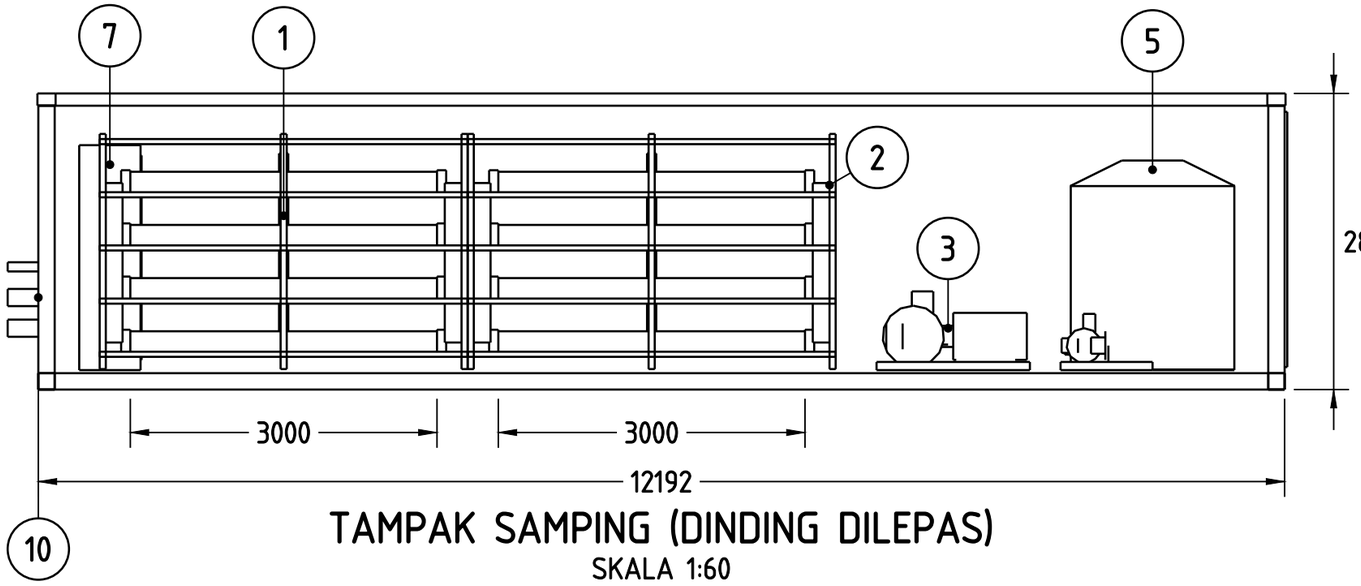
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Equipment Drawing			Rev: 0
	Author : NAMA TIM	Reaktor AnMBR Standar R-201 s/d R-214	Unit: mm	Scale: 1:80 / 1:60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-V-201			Sheet number: 2 / 8



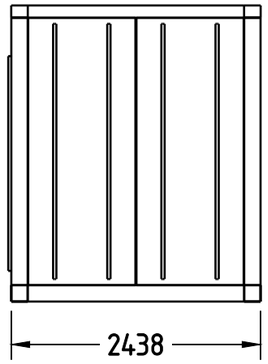
DENAH (ATAP DILEPAS)
SKALA 1:60



ISOMETRI POTONG (TANPA SKALA)



TAMPAK SAMPING (DINDING DILEPAS)
SKALA 1:60

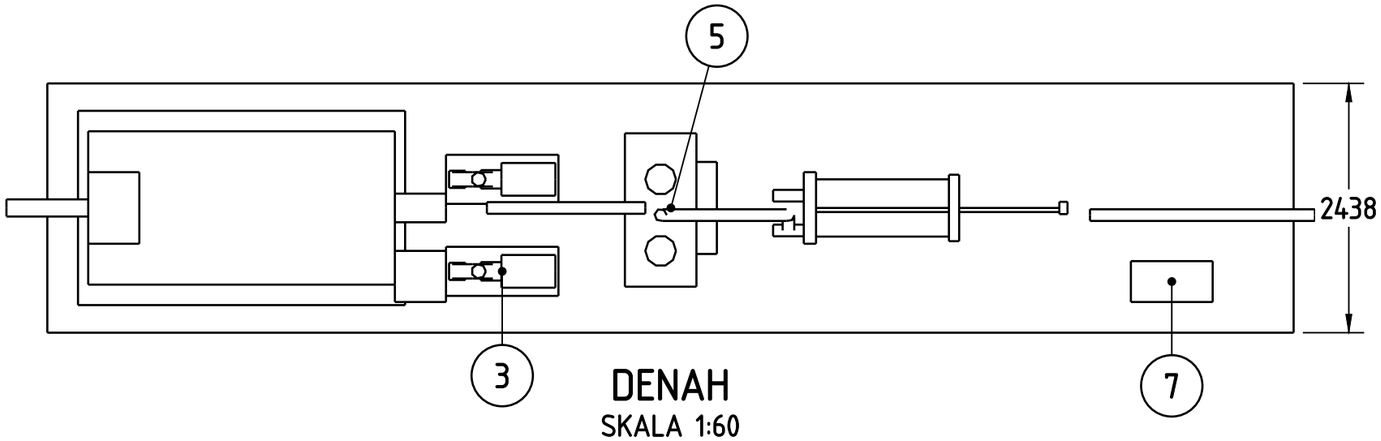


TAMPAK UJUNG
SKALA 1:60

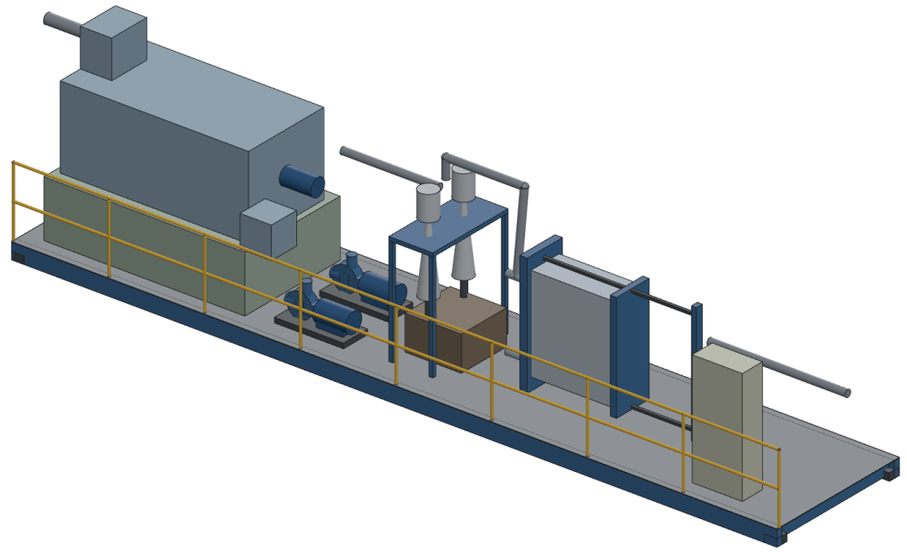
CATATAN :
1. Modul 2 bank x 5 kolom x 4 tingkat (40 posisi), jarak antar tingkat 520 mm.
2. Fluks rancangan 15 LMH, luas total 996 m².
3. Massa angkut ± 16 t; kontainer ISO 40 ft HC.
4. Lorong servis 850 mm di sisi pompa.
5. Dinding kiri & atap dilepas pada tampak / denah untuk kejelasan.

DAFTAR BAGIAN			
No	Tag	Uraian	Jml
1	MB-301	Modul UF tubular Ø220 x 3000, 30 m²	34 (+6 cad.)
2	–	Rak modul baja galvanis 2 bank	2
3	P-301A/B	Pompa cross-flow 149 m³/jam, 30 kW	2
4	P-302	Pompa permeat, VSD	1
5	TK-302	Tangki CIP Ø1600 x 1800, 4 m³	1
6	P-304	Pompa CIP	1
7	–	Panel MCC + PLC membran	1
8	–	Header umpan / retentat DN150	4
9	–	Header permeat DN80	2
10	–	Nosel dinding ke reaktor (DN150/80)	3
11	–	Kontainer ISO 40 ft HC, kisi ventilasi + pintu samping	1

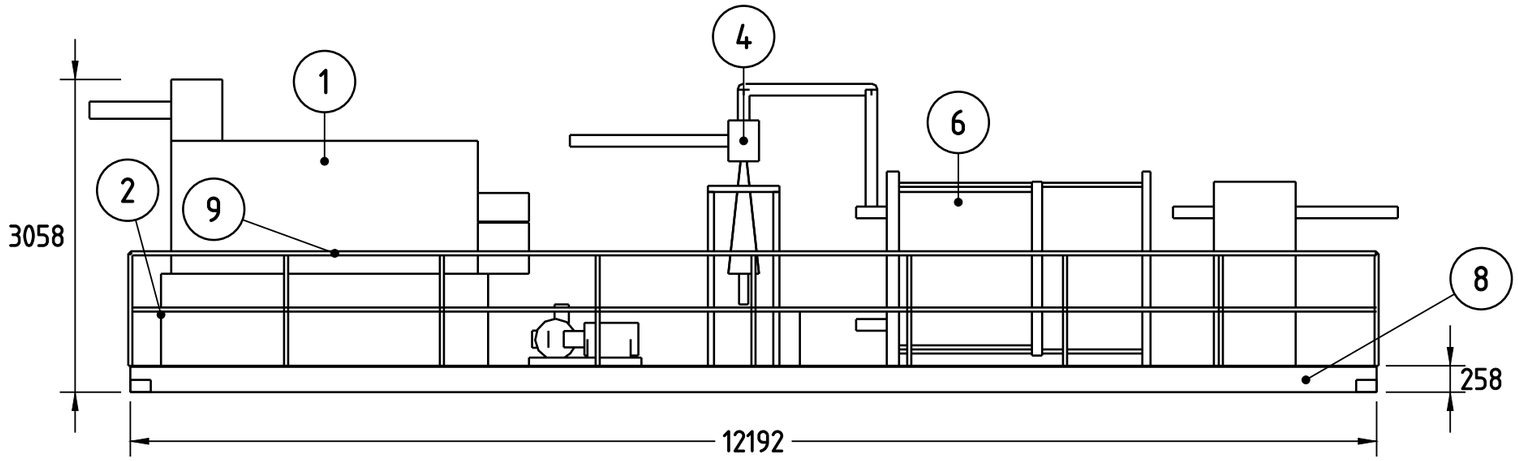
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	PM-4 Kontainer Membran UF (MB-301, P-301, P-302, TK-302)	Unit: mm	Scale: 1 : 60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-004			Sheet number: 3 / 8



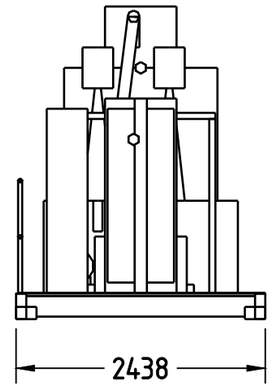
DENAH
SKALA 1:60



ISOMETRI (TANPA SKALA)



TAMPAK DEPAN
SKALA 1:60

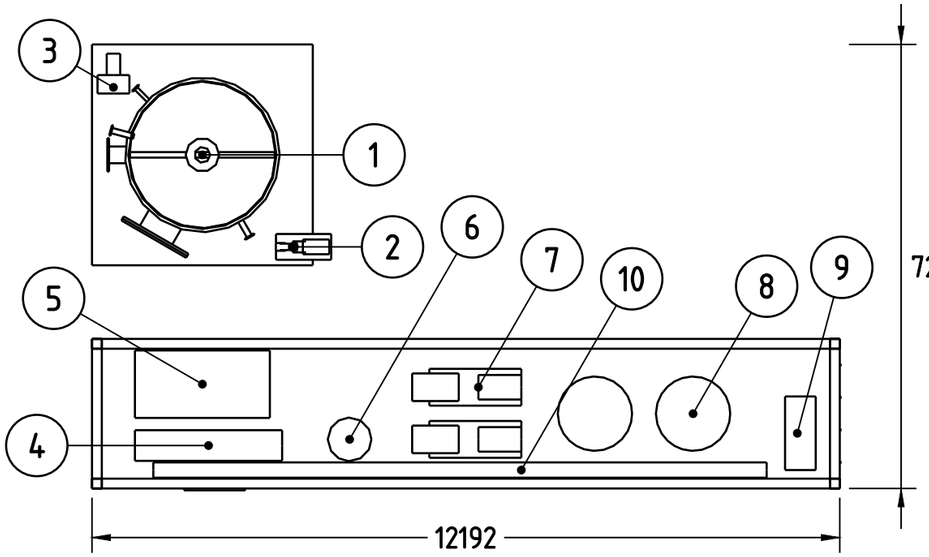


TAMPAK UJUNG
SKALA 1:60

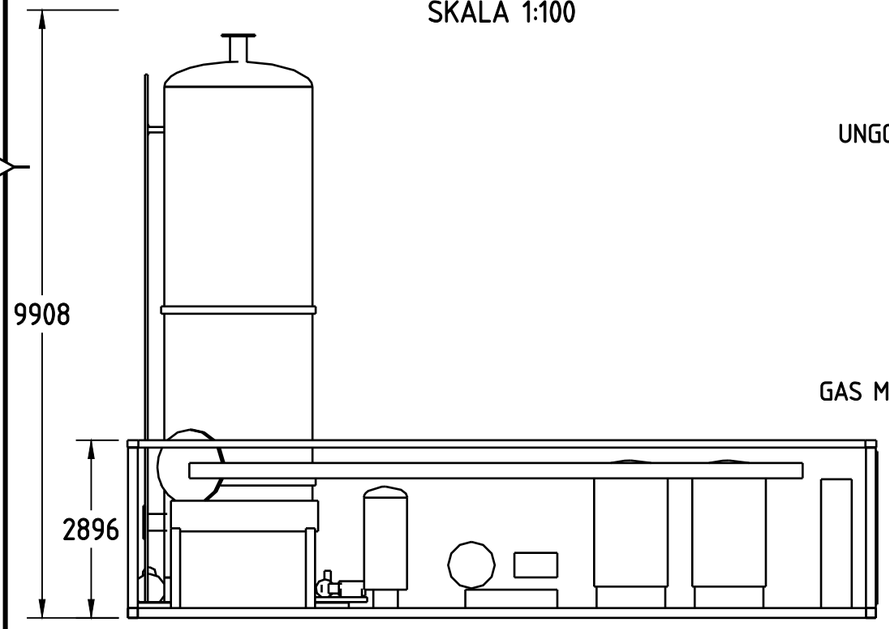
CATATAN :
1. Seluruh alat dirakit & diuji di bengkel.
2. Pengiriman: 1 truk trailer 40 ft, ± 12 t.
3. Sambungan: POME masuk DN150 (kiri), POME keluar DN100 ke TK-101 (kanan).

DAFTAR BAGIAN			
No	Tag	Uraian	Jml
1	SC-101	Saringan drum putar Ø900 x 2600, celah 0,75 mm, 24.4 m³/jam	1
2	–	Bak sump 3 m³ di bawah saringan	1
3	P-101A/B	Pompa umpan desander, H 20 m, VSD	2
4	HC-101	Hidrosiklon desander 2 x Ø300	2
5	–	Bak pasir (grit bin) 0,4 m³	1
6	E-101	Pendingin POME pelat 812 kWth, SS316	1
7	–	Panel kontrol lokal	1
8	–	Skid baja UNP 250, 12192 x 2438	1
9	–	Pegangan tangan Ø42,4	1

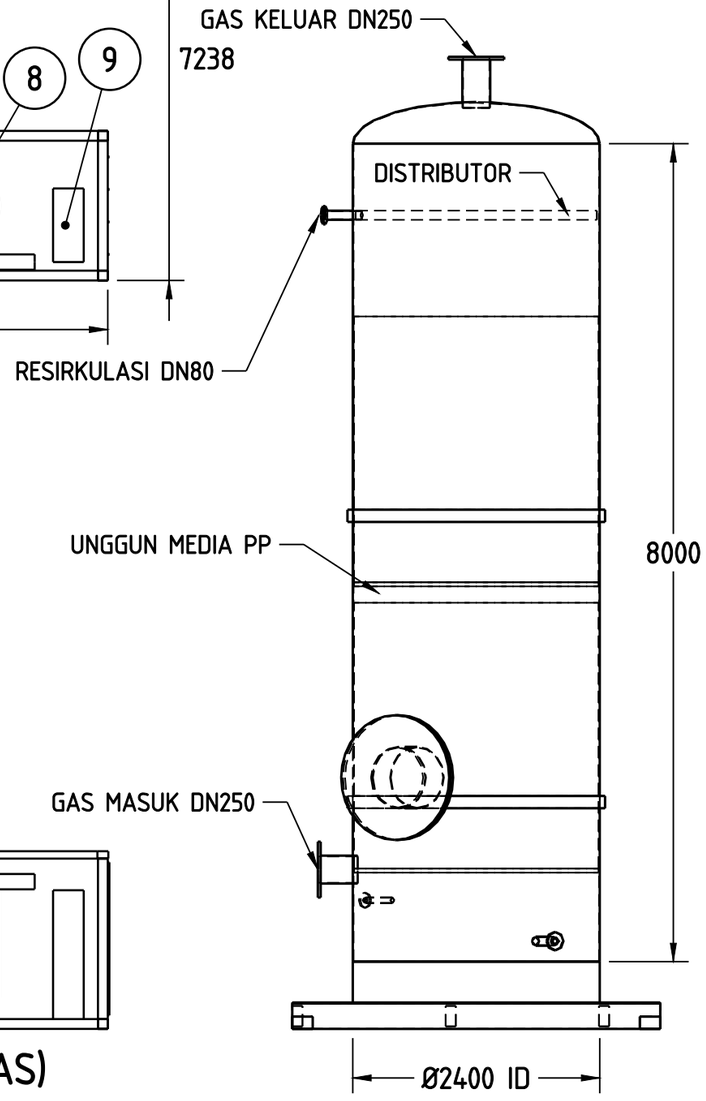
University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	PM-1 Skid Praolah POME (SC-101, HC-101, E-101)	Unit: mm	Scale: 1 : 60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-001			Sheet number: 4 / 8



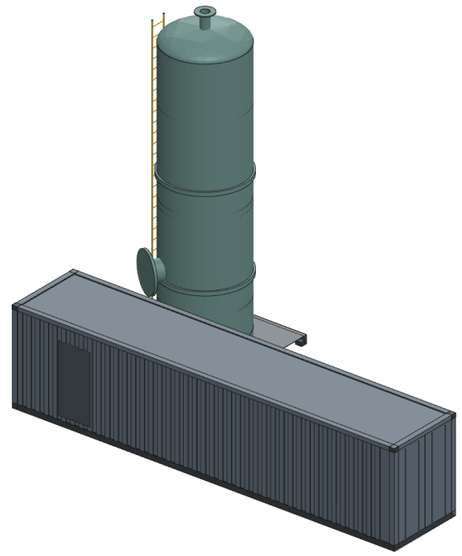
DENAH (ATAP DILEPAS)
SKALA 1:100



TAMPAK DEPAN (DINDING DILEPAS)
SKALA 1:100



BTF-301 (ELEVASI)
SKALA 1:60

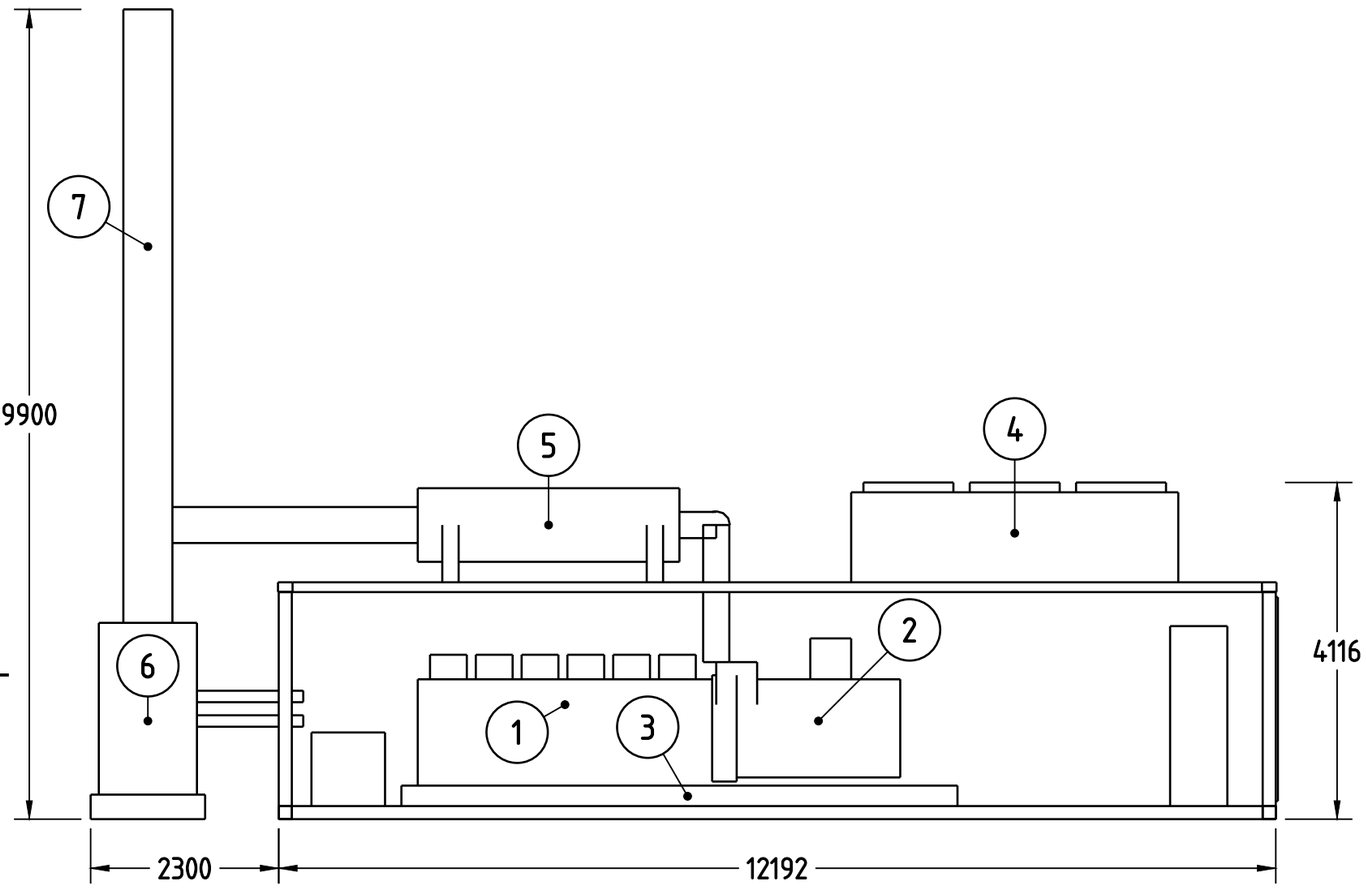


ISOMETRI (TANPA SKALA)

DAFTAR BAGIAN			
No	Tag	Uraian	Jml
1	BTF-301	Bio-trickling filter Ø2400 x 8000, media PP 16.4 m³	1
2	P-305	Pompa resirkulasi nutrien BTF	1
3	K-301	Blower udara BTF 5.6 Nm³/jam	1
4	HX-301	Pendingin gas shell-tube, 15.3 kWth	1
5	CH-301	Chiller paket berpendingin udara	1
6	V-301	KO pot Ø700 + demister	1
7	K-302A/B	Blower biogas 537 m³/jam, ΔP 20 kPa	2
8	AC-301A/B	Filter karbon aktif Ø1200, 2 bed	2
9	–	Panel kontrol + analis gas	1
10	–	Header gas DN250	1
11	–	Kontainer ISO 40 ft HC (ATEX zona 2)	1
12	–	Skid BTF 3600 x 3600 + tangga	1

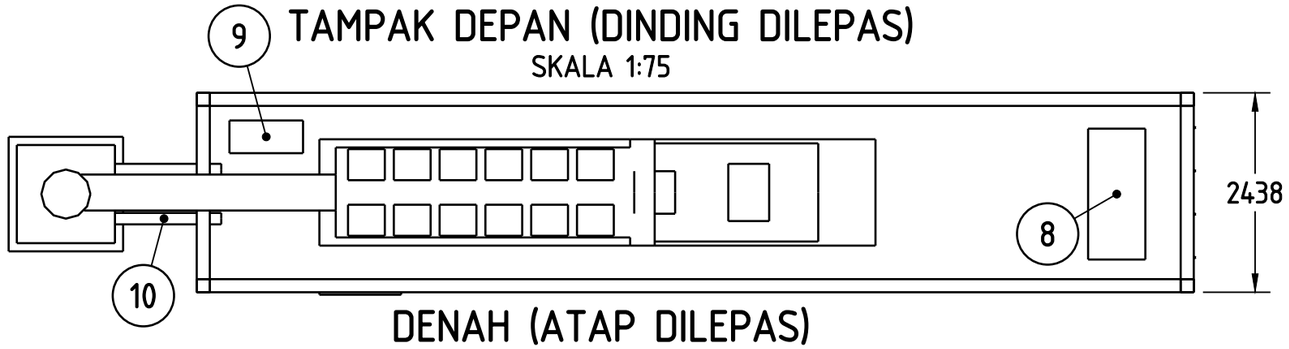
CATATAN :
1. BTF dikirim sebagai bejana terpisah di trailer.
2. H_2S: 2887 ppm masuk → 34 ppm ke genset.
3. Kontainer: ventilasi paksa + detektor gas.

University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	PM-6 Skid Pengolahan Biogas (BTF-301, K-302, AC-301)	Unit: mm	Scale: 1:100 / 1:60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-006			Sheet number: 5 / 8



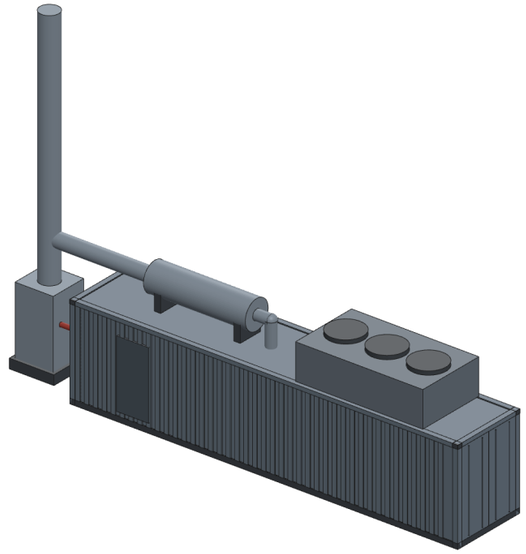
TAMPAK DEPAN (DINDING DILEPAS)

SKALA 1:75



DENAH (ATAP DILEPAS)

SKALA 1:75



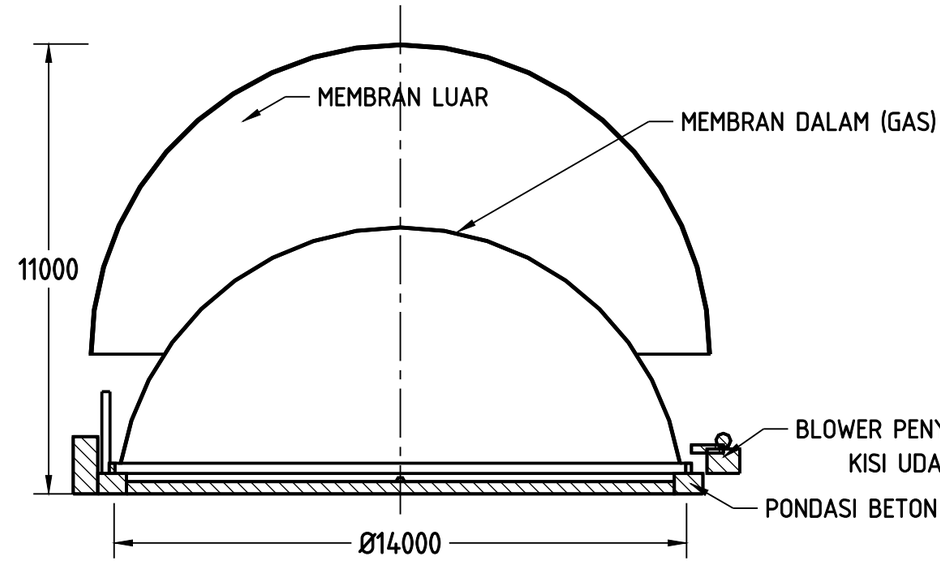
ISOMETRI (TANPA SKALA)

DAFTAR BAGIAN (PER UNIT)			
No	Tag	Uraian	Jml
1	GE-401	Mesin gas V-12, 800 kWe, ηel 39.5 %	1
2	–	Generator sinkron 400 V, 50 Hz	1
3	–	Rangka dasar + peredam getar	1
4	–	Radiator / dump cooler 3 kipas	1
5	–	Silencer gas buang Ø900 x 3200	1
6	HX-401	Pemulih panas gas buang + jaket, 1439 kWth	1
7	ST-401	Cerobong Ø600, tinggi 9,9 m	1
8	–	Panel kontrol + sinkronisasi	1
9	–	Katup gas train + pengatur tekanan	1
10	–	Pipa air panas HWS/HWR DN80	2
11	–	Kontainer ISO 40 ft HC, kedap suara 75 dB(A) @ 7 m	1

CATATAN :

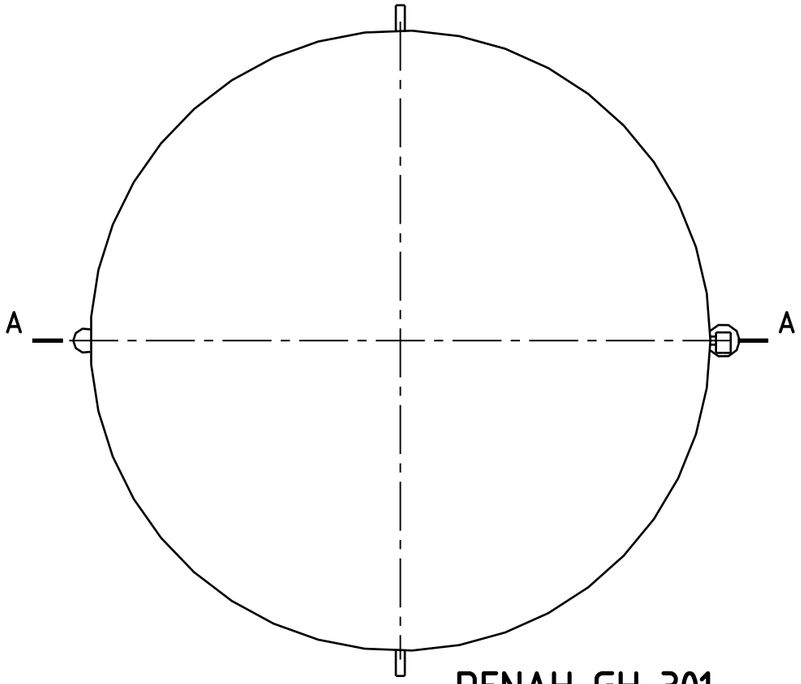
1. Dua unit identik GE-401 / GE-402 (2 kontainer).
2. Bahan bakar: biogas H_2S ≤ 50 ppm dari PM-6.
3. HX-401 & ST-401 dikirim sebagai skid PM-9.
4. Massa angkut ± 24 t per kontainer.

University: UNIVERSITAS	Checked by : –	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	PM-8 Kontainer CHP 800 kWe + HX-401 / ST-401 (PM-9)	Unit: mm	Scale: 1 : 75	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : –		BG-M-008			Sheet number: 6 / 8



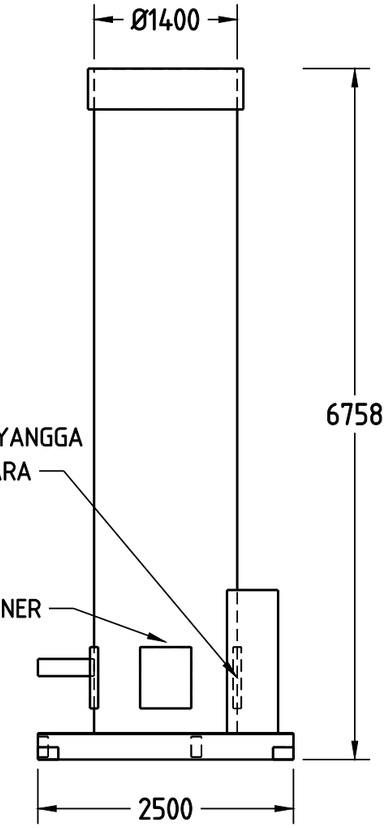
POTONGAN A-A GH-301

SKALA 1:150



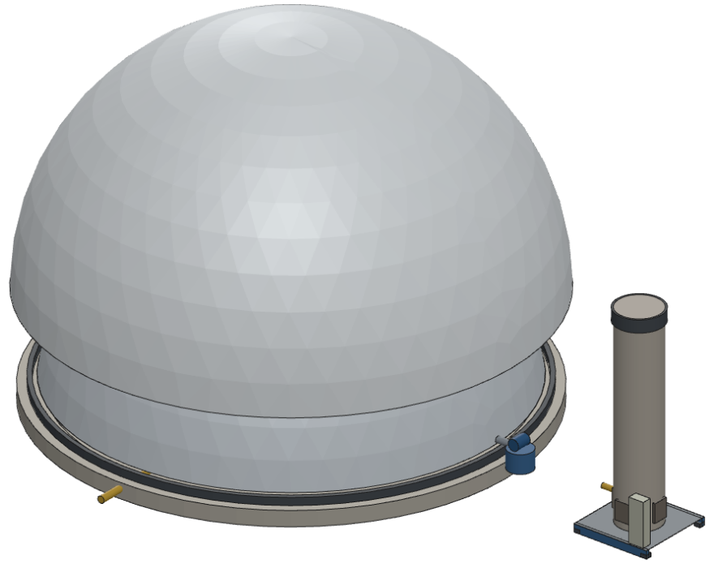
DENAH GH-301

SKALA 1:150



FL-301 (ELEVASI)

SKALA 1:60



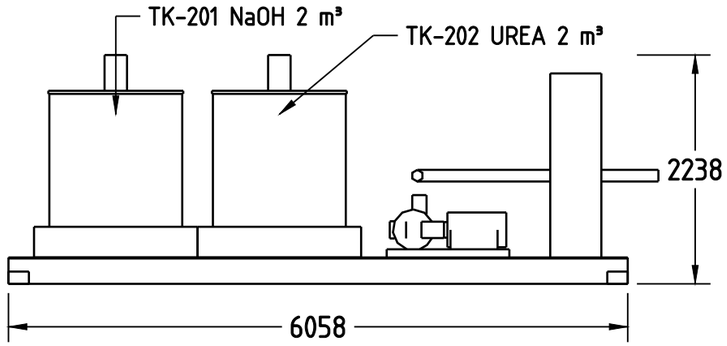
ISOMETRI (TANPA SKALA)

DATA DESAIN		
Tag	Parameter	Nilai
GH-301	Volume gas berguna	1135 m³ (± 2 jam P95)
	Volume kubah total	1414 m³, Ø14.0 x 10.5 m
	Membran luar / dalam	PVC-poliester 1100 g/m² / PE kedap metana
	Tekanan operasi	2 – 3 mbar(g), blower udara penyangga
	Pondasi	Cincin beton Ø14800, lantai beton 300 mm
	Pengiriman	membran terlipat di 1 kontainer 20 ft
FL-301	Kapasitas	624 Nm³/jam, suhu nyala ≥ 1000 °C
	Tipe	Flare tertutup, pilot otomatis, skid 2,5 x 2,5 m

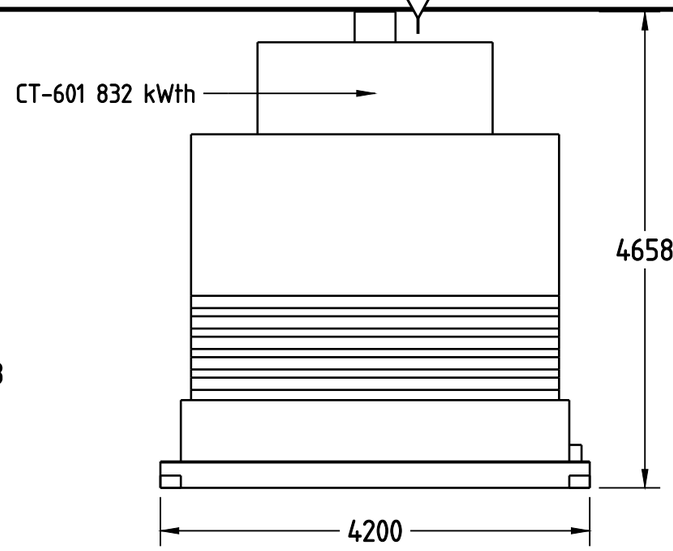
CATATAN :

1. Membran luar dijaga tegang oleh blower udara penyangga.
2. Membran dalam naik-turun sesuai isi gas (LIC-401).
3. Flare menyala otomatis saat GE berhenti / GH penuh (PIC-407).
4. Jarak aman GH ke flare ≥ 15 m (lihat tata letak BG-GA-001).

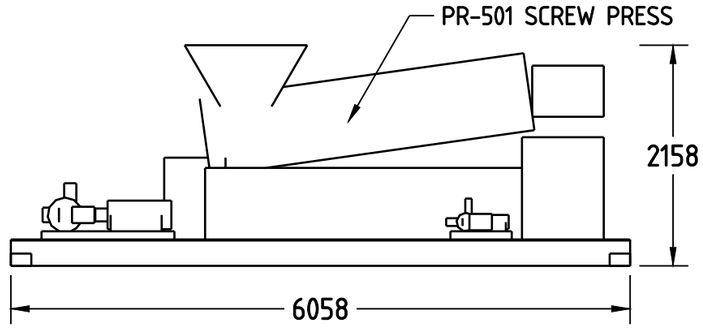
University: UNIVERSITAS	Checked by : -	Design Lab.	Document type: Assembly Drawing			Rev: 0
	Author : NAMA TIM	PM-7 Penampung Gas GH-301 + Flare Tertutup FL-301	Unit: mm	Scale: 1:150 / 1:60	Creation date: 15/10/2026	Size: A3
	Eng. Approved by : -		BG-M-007			Sheet number: 7 / 8



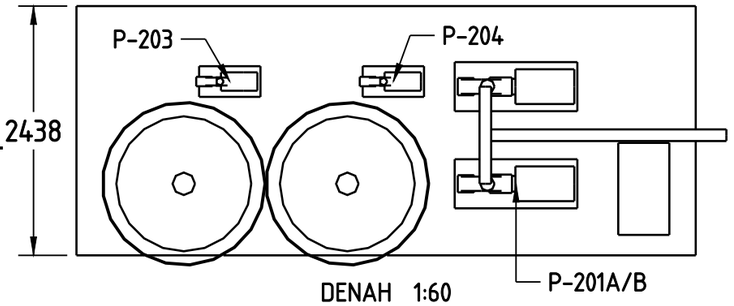
PM-5 TAMPAK DEPAN 1:60



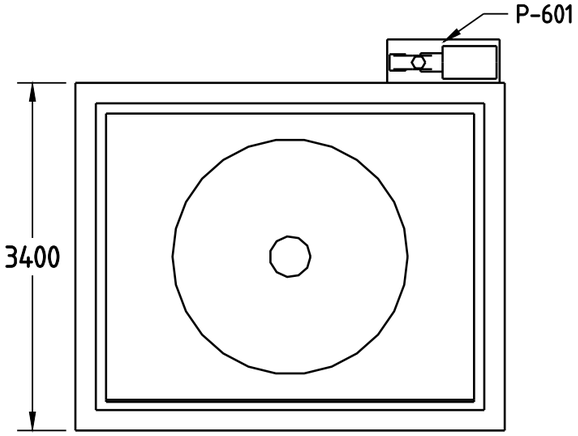
PM-9 TAMPAK DEPAN 1:60



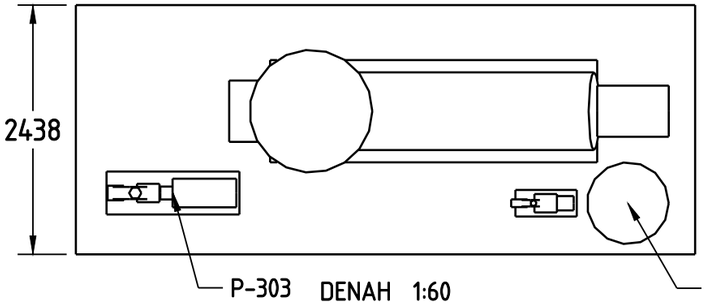
PM-10 TAMPAK DEPAN 1:60



DENAH 1:60

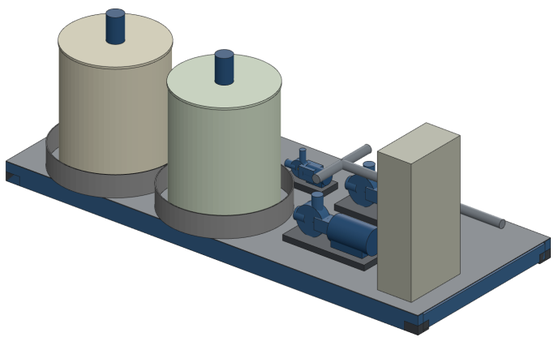


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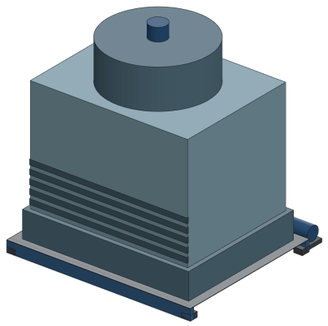


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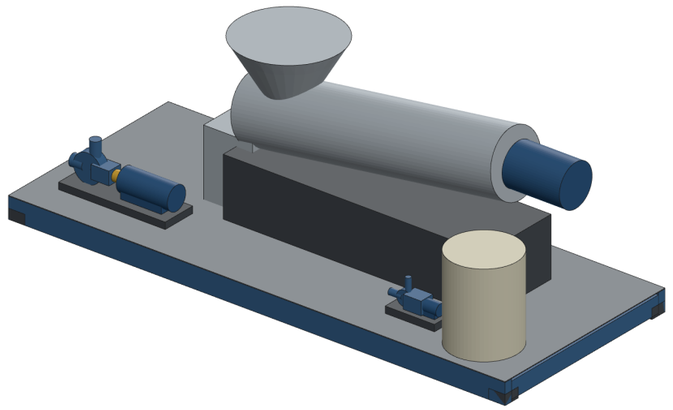
ISOMETRI PM-10



ISOMETRI PM-5



ISOMETRI PM-9



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